Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **OWLabs**
b. Team leader name: **Chirag HS**
c. Problem Statement: **Manufacturing: Intelligent Operations & Industrial Efficiency**

Brief about the idea

TwinForge AI

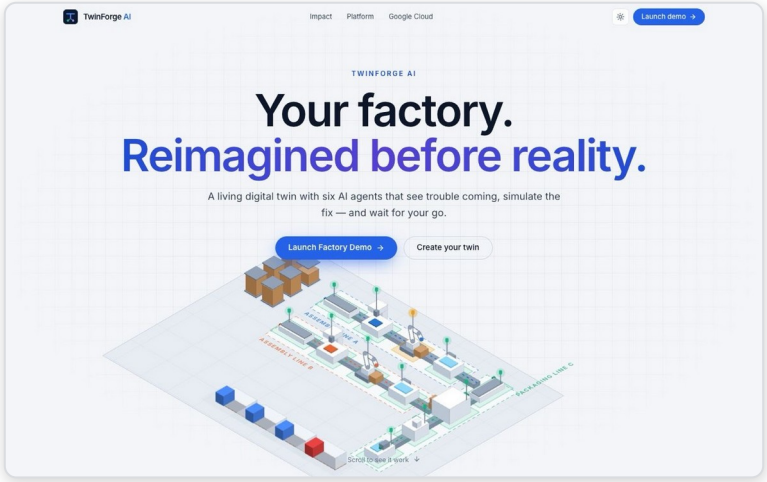
Your factory. Reimagined before reality.

An agentic industrial digital twin on Google Cloud. It mirrors a factory as a live model, spots trouble early, simulates the alternatives and proposes a plan, which a person approves before anything changes.

See
A live twin of lines, machines, materials, orders, cameras and mobile robots, in topology, isometric, floor-plan and 3D views.

Understand
Six specialist AI agents investigate incidents; a multi-agent copilot built with Google's ADK answers with traceable figures.

Decide
A deterministic plant engine simulates the options; a person approves, the twin executes, and every step is audited.



OUR PRINCIPLE

Language models explain, extract and orchestrate. Numbers come from code. Decisions come from people.

Opportunities

How is it different?

- Not a monitoring dashboard: the twin simulates forward, so options are compared before the plant changes.
- Not a chatbot that guesses: agents compute every figure on a deterministic plant engine and show the evidence.
- Not months of modelling: the Twin Builder drafts a twin from a drawing and a plain-language description.

How will it solve the problem?

- Today, health, quality, materials and orders sit in separate systems, and comparing options by hand takes hours.
- TwinForge joins them in one live twin, predicts trouble early and ranks plans on delivery, cost, risk and sustainability.
- Demo crisis: the recommended plan avoids about 700 late units at about a quarter of the cost of carrying on (model estimates).

USP of the solution

- Models explain, code computes, people decide: every figure the AI states is checked against computed results.
- A person approves every plan, with rollback and a full audit trail; there is no control path to equipment.
- Open by design: the same 12 tools are served over MCP, so other agents can use the twin, and still only propose.

List of features offered by the solution

Command Center

OEE, output, machine health, energy, scrap, shortages and delivery risk on one screen.

AI Operations

Six specialist agents stream findings with evidence, from observe to recommend.

What-If Simulator

Compare plans on more than ten levers; results recalculate instantly in the browser.

Digital Twin Studio

Topology, isometric, floor-plan and real-time 3D views of one live twin.

Approvals and audit

A person approves, the twin executes, and every step is recorded and reversible.

Visual Quality and CCTV

FPY, DPMO and defect Pareto, nine cameras, and AI visual checks a person confirms.

Twin Builder

From a floor plan and a description to a running twin, with provenance on every element.

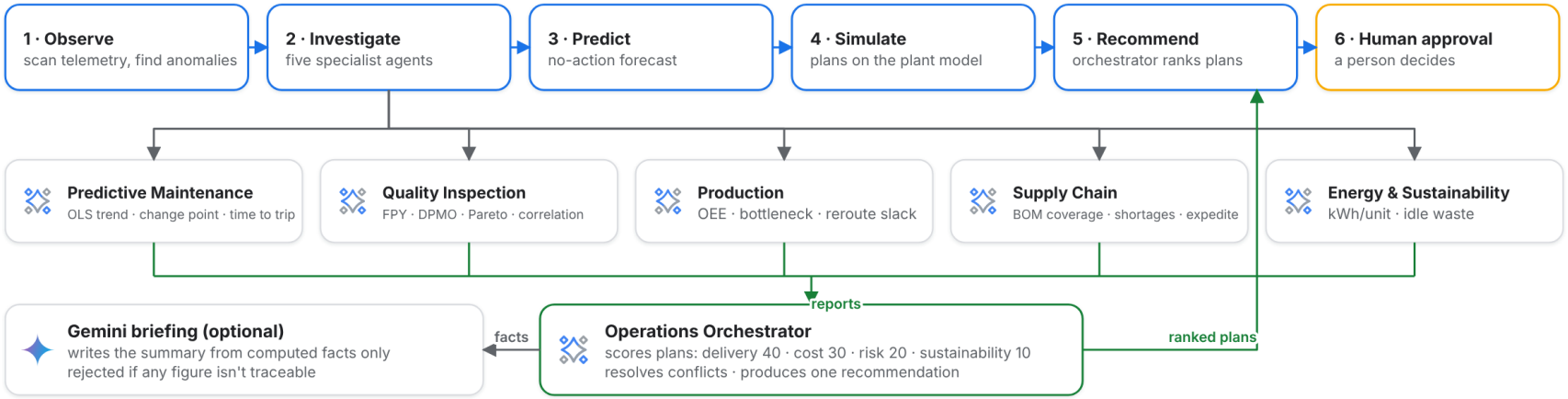
Playground

Drag machines and robots onto the floor, set automation rules, preview in 3D and apply.

Copilot on ADK + MCP

Multi-agent copilot with visible tool calls; the same 12 tools are served over MCP.

Process flow diagram or Use-case diagram

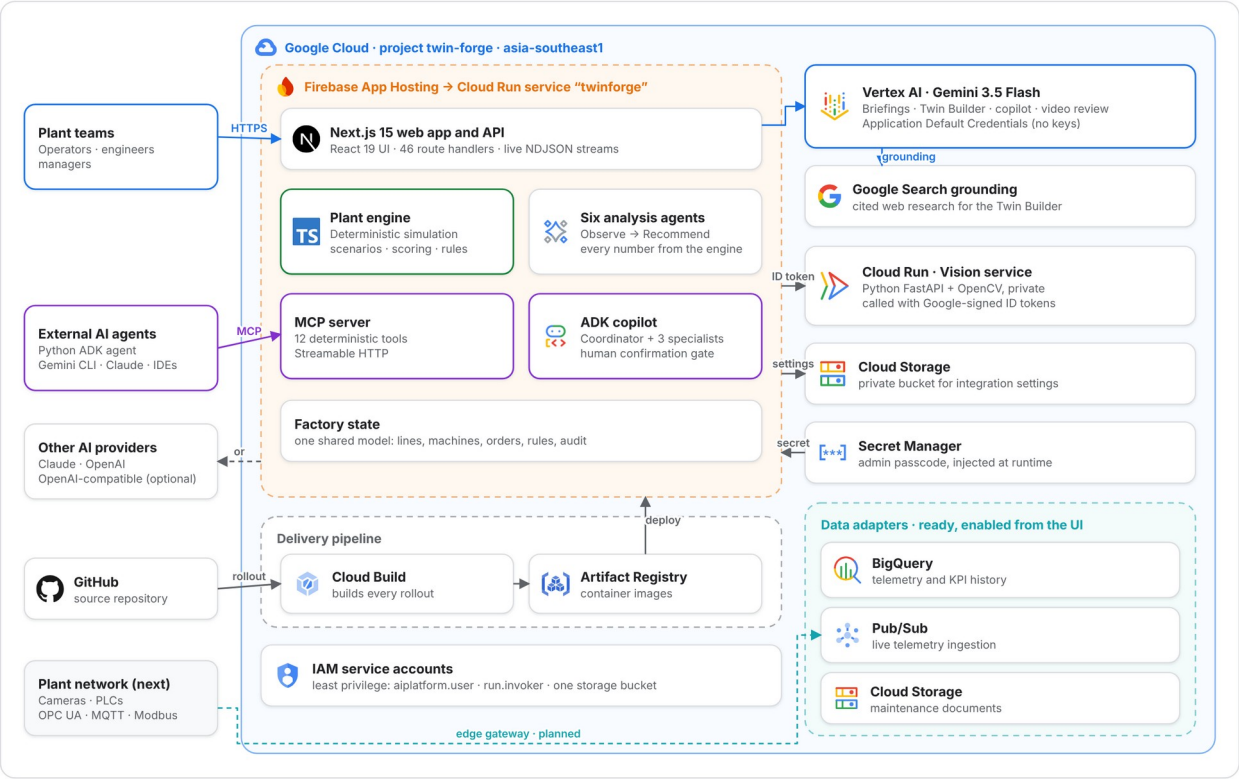


Every finding lists its inputs, method, evidence and confidence. The workflow runs on a frozen snapshot and streams each stage live.
Agents can only propose: the recommendation waits in Approvals until a person approves, rejects or rolls it back.

After approval

Execute the twin runs the plan as directives over simulated time → **Audit** every step recorded with links; any step can be rolled back

Architecture diagram of the proposed solution



One service

Next.js on Firebase App Hosting (Cloud Run) runs the UI, API, plant engine, ADK copilot and MCP server.

Gemini on Vertex AI

Gemini 3.5 Flash through Application Default Credentials, so no API keys.

Private vision

FastAPI + OpenCV on Cloud Run, called with Google-signed ID tokens.

Secure by default

Secret Manager, a private Cloud Storage bucket and least-privilege IAM.

Ready to scale

BigQuery and Pub/Sub adapters built in; runs in asia-southeast1.

Technologies to be used in the solution

AI and agents

- Vertex AI Gemini 3.5 Flash
- Google Gen AI SDK
- Agent Development Kit TypeScript + Python
- Google Search grounding
- Model Context Protocol

Google Cloud

- Firebase App Hosting
- Cloud Run
- Cloud Build
- Artifact Registry
- Secret Manager
- Cloud Storage
- IAM
- BigQuery
- Pub/Sub

Web application

- Next.js
- React
- TypeScript
- Tailwind CSS
- Zustand
- Zod
- React Flow
- Recharts
- Radix UI
- Node.js

3D and media

- three.js
- React Three Fiber
- drei
- postprocessing
- hls.js

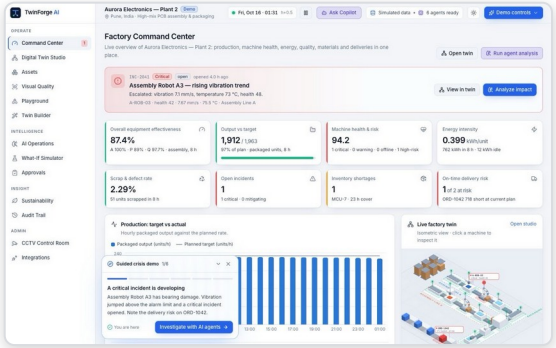
Python services

- Python
- FastAPI
- OpenCV
- google-adk[mcp]

Testing

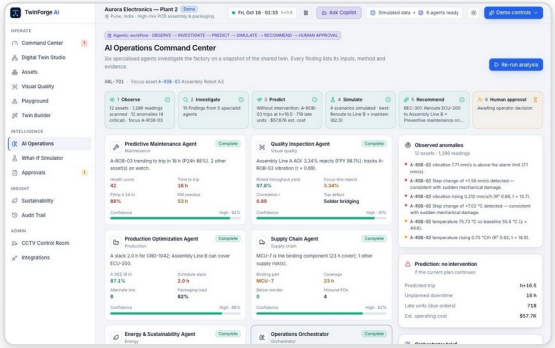
- Vitest
- Playwright
- pytest
- GitHub

Snapshots of the prototype (1/2)



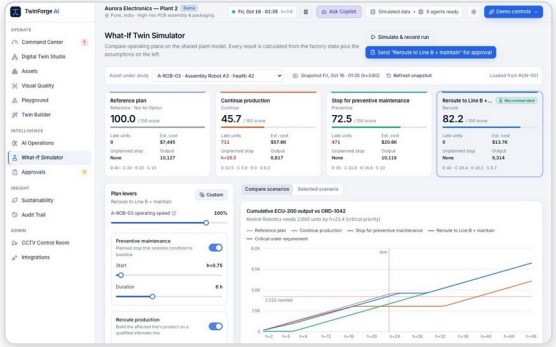
Command Center

The whole plant on one screen during the crisis, with the guided demo panel.



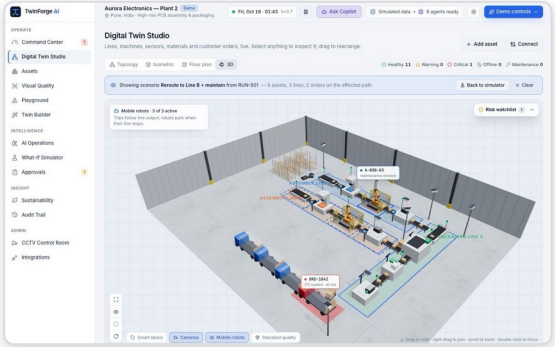
AI Operations

Six agents stream findings; a trip is predicted about 16 hours ahead.



What-If Simulator

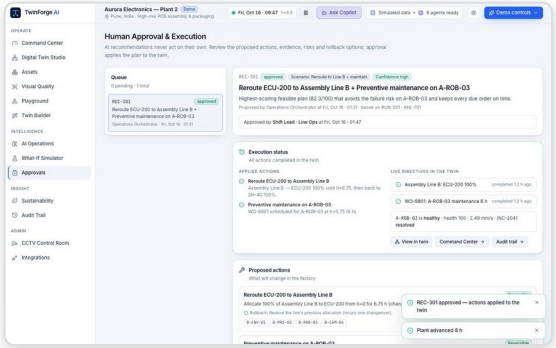
Four plans compared; reroute to Line B + maintain scores 82.2 with no late units.



Digital Twin in 3D

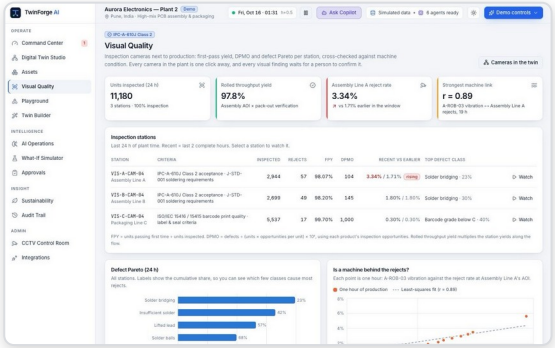
Orbit the live plant; robots animate and the recommended path is highlighted.

Snapshots of the prototype (2/2)



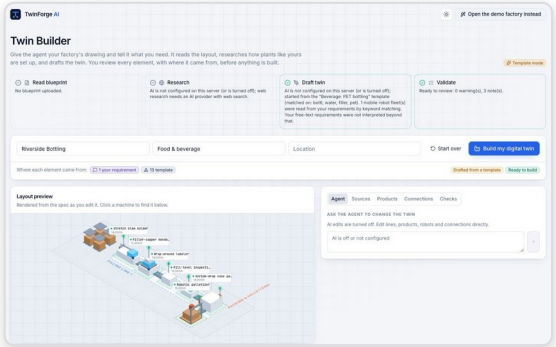
Approvals

The approved plan ran: Line B took over and the failing robot is healthy again.



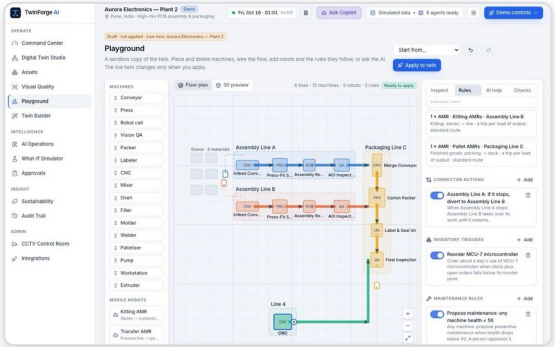
Visual Quality

FPY, DPMO and defect Pareto; robot vibration tracks Line A rejects (r = 0.89).



Twin Builder

Review screen with stages, provenance counts and a live layout preview.



Playground

Place machines and robots on the floor and set automation rules.

Prototype Performance report/Benchmarking

Benchmark: plans compared on the twin in the demo crisis

model estimates on demo data

Plan	Score / 100	Late units	Estimated cost	Unplanned stop
Reference plan (asset assumed healthy)	100.0	0	$7.4K	None
Continue production	45.7	711	$57.6K	Trip at about h+16.5
Stop for preventive maintenance	72.5	471	$20.6K	None
Reroute to Line B + maintain (recommended)	**82.2**	**0**	**$13.7K**	**None**

Scores weight delivery 40, cost 30, risk 20 and sustainability 10 against the reference plan.

Measured on the live prototype

≈ 5.5 s
Gemini agent briefing

8–23 s
Copilot answer with tool calls

≈ 15 s
External Python ADK agent, 4 tool calls

≈ 78 s
Twin Builder draft with web research

Instant
What-If recalculation, in the browser

19 / 19
Copilot figures traced to tool results

30×
Live plant speed against real time

149
Automated tests: 117 unit, 16 E2E, 16 Python

Additional Details/Future Development (if any)

Current scope

- Runs on a modelled plant with sample data, so results are estimates.
- One factory per deployment; state is held per instance.
- Live telemetry arrives through Pub/Sub today; OPC UA, Modbus and MQTT need an edge gateway.
- Camera footage in the demo is licensed video of other facilities.

What comes next

Multi-site
Firestore shared state, Firebase Auth sign-in and several factories per tenant.

Plant connectivity
Edge gateway for OPC UA, Modbus and MQTT into Pub/Sub; camera analytics at the edge.

Server-side clock
Cloud Scheduler drives the plant clock, with no instance cap.

Smarter Twin Builder
A self-correction loop on ADK; Agent Engine hosting for the Python agent.

Learned inspection
Trained inspection models, validated with a measurement-system study before use.

Agent-to-agent
An A2A endpoint so enterprise agents can delegate work to TwinForge.

Provide links to your:



GitHub Public Repository

https://github.com/The-OWLabs/twin-forge-ai



Demo Video Link (3 Minutes)

Add the public YouTube, Vimeo or Google Drive link before submitting



Final Product Link

https://twinforge--twin-forge.asia-southeast1.hosted.app

Google Cloud H2S

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Thank you

